

AHB-Lite v1.0 System Design and Verification Plan

1. Supported and Unsupported Features

Supported Features

- **Address Signals (`HADDR`)**: Fully supported for all memory accesses.
- **Transfers & Bursts**: Supports Single transfers and bursts of at least 1 transition increment minimum (e.g., `SINGLE` , `INCR`).
- **Transfer Sizes**: Word (32-bit), Halfword (16-bit), and Byte (8-bit) access with correct byte-lane masking.
- **Wait States**: Full support for zero-wait state operations and slave-inserted wait states via `HREADY` .

Unsupported Features

- **Error Handling (`HRESP`)**: The FSM logic is fully implemented to handle 2-cycle `HRESP=ERROR` responses and abort bursts, but this feature is considered **unsupported**.
- **Protection (`HPROT`)**: Implemented in the master interface and tied to a default value (Data + Privileged), but dynamically driving this signal is **not supported**.
- **Locked Transfers (`HMASTLOCK`)**: Active arbiter locking is **not supported**.
- **Unaligned Transfers**: Not supported. The master expects addresses to be aligned to the transfer size.
- **Narrow Bursts**: Data bus width is fixed to 32 bits.

2. System Design Overview

The AHB-Lite system is designed as a modular, synthesizable AMBA 3 AHB-Lite compliant environment. It consists of a pipelined Master, an Interconnect (Decoder + Mux), and multiple Slaves.

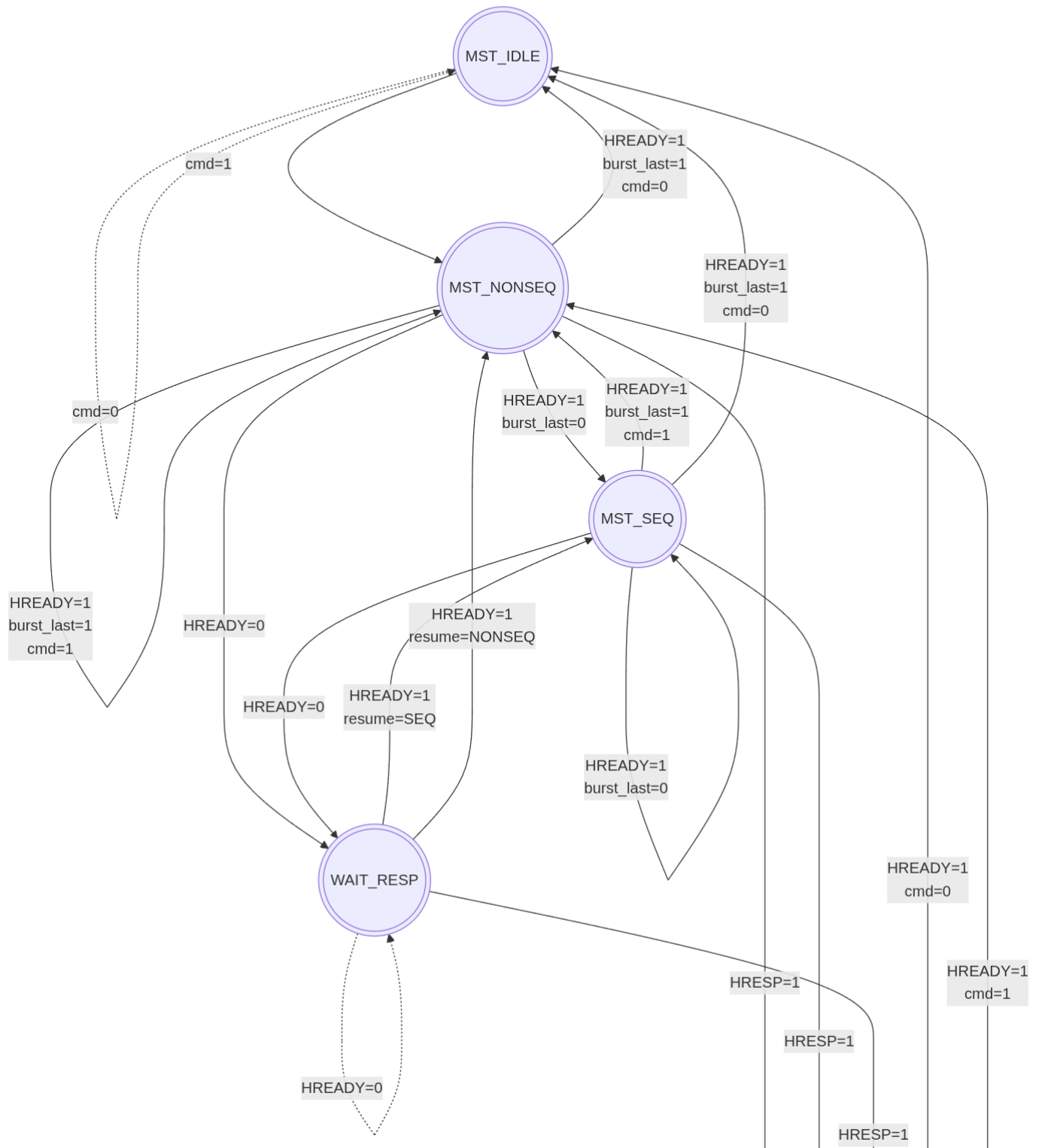
1.1 Architecture

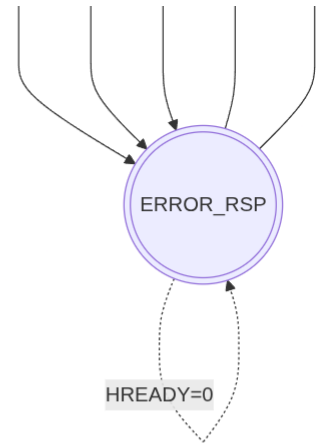
- **AHB Master (`ahb_lite_master`)**: Initiates transfers. It is highly decoupled into three sub-modules:
 - **FSM (`ahb_master_fsm`)**: Controls the `HTRANS` state and tracks the address/data phases.
 - **Address Generator (`ahb_addr_gen`)**: Calculates sequential and wrapping burst addresses dynamically based on `HSIZE` and `HBURST` .
 - **Burst Counter (`ahb_burst_counter`)**: Tracks remaining beats for fixed-length bursts.
- **Interconnect**:
 - **Decoder (`ahb_decoder`)**: Purely combinational. Decodes `HADDR[11:10]` to generate one-hot `HSEL` signals for up to 3 slaves.
 - **Multiplexer (`ahb_mux`)**: Registers `HSEL` during the address phase to correctly route the selected slave's `HRDATA` , `HREADYOUT` , and `HRESP` during the data phase.
- **Slaves**:
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- **Slave Memory (ahb_slave_mem)**: A 1KB memory slave that uses global HREADY to properly latch address-phase control signals (HADDR , HSIZE , HWRITE) for execution in the data phase. It performs byte-lane masking for halfword and byte accesses.
- **Slave BFM (ahb_slave_bfm)**: A Bus Functional Model capable of injecting wait states (driving HREADYOUT low) and generating 2-cycle Error responses.

2.2 Master FSM Design

The Master FSM operates with 5 primary states to manage the pipelined nature of AHB:





MST_IDLE : Default state. Drives **HTRANS_IDLE** . Transitions to **NONSEQ** upon a new command.

MST_NONSEQ : First beat of a transfer. Drives **HTRANS_NONSEQ** .

MST_SEQ : Subsequent beats of a burst. Drives **HTRANS_SEQ** .

MST_WAIT_RESP : Entered if a slave pulls **HREADY** low at the end of an address phase. Holds the current **HTRANS** and **HADDR** until **HREADY** goes high.

MST_ERROR_RSP : Handles the 2-cycle error response mechanism when **HRESP** is asserted.

2.3 Data Pipelining Strategy

To align write data (**HWDATA**) with the data phase (which trails the address phase by one cycle), the Master utilizes a 2-stage holding pipeline:

- **wdata_hold_r** : Latches the testbench's write data at the end of the address phase.
- **hwdatar** : Updates to **wdata_hold_r** at the start of the data phase, holding **HWDATA** stable for the duration of the data phase, even if extended by wait states.

3. Verification Plan

The verification strategy employs a direct-test approach using a custom SystemVerilog testbench (**tb_ahb_lite_system**). The goal is to stimulate all edge cases of the AHB-Lite protocol.

3.1 Test Sequences

The test suite consists of 15 targeted tests designed to stress both basic features and edge cases.

Test Category	Description	Goal
Basic Transfers	Single Write, Single Read, Write-Read-Verify	Validate basic data integrity and sim
Pipelining	Back-to-Back Writes and Reads	Verify HTRANS transitioning correctly and pipelined
Bursts (INCR)	INCR4 , INCR8 , INCR16 sequences	Ensure sequential address generatic
Bursts (WRAP)	WRAP4 burst crossing boundary	Verify precise bitwise address maski → 0x3C → 0x30).
Wait States	Random HREADYOUT delay injection	Confirm master stalls properly in MS holding outputs stable.
Sub-word Access	Byte (HSIZE_BYTE) and Halfword accesses	Verify slave byte-lane selection logic
Error Response	2-cycle HRESP=1 injection	Test FSM ability to immediately abor
Reset Recovery	Assert HRESETn mid-transfer	Check asynchronous recovery direct